Applied Microeconomics — Master Notes

by Jakob V. Stangel

> One running document for the whole course, organised **by lecture (= textbook chapter)**. Format per lecture: **Résumé → ★ Heart of it → Key concepts → Section-by-section → Cases → Key terms → Exam pointers.** Textbook: **Richard Friberg, *A Free Introduction to Microeconomics*** · Lecturer: **Luigi Butera** (lbu.eco@cbs.dk). *Exam: written sit-in, **open book, No AI**, Week 50 (9 Dec 2026).*

Overview — where everything is

(Everything below is clickable.)

Lecture 1: Principles of Economics (Chapter 1)

Required reading: Friberg, ch. 1 (*Introduction*).

Theme of the lecture (Butera): the "**economic way of thinking**" — economics studies how people **allocate scarce resources** among competing needs, using **three principles** (optimization · equilibrium · empiricism) and **models** (simplified, testable) to understand and predict behaviour.

> ★ **The heart of the chapter:** economics = **doing the best you can with scarce resources.** The toolkit is **three principles — Optimization · Equilibrium · Empiricism**; the two workhorses of optimization are **opportunity cost** and **marginal thinking**; and the method is building **positive (testable) models**, not making normative judgments. "*All models are useful, but all models are wrong.*"

The red thread. Everything starts from **scarcity** — without it we wouldn't need economics. From scarcity flows: you must **choose** (optimize) → every choice has an **opportunity cost** and is decided **at the margin** → many people choosing interact until no one wants to move (**equilibrium**) → and we test our claims about all this with **models + data** (empiricism). For any decision ask: *what's scarce · what's the best alternative given up (opportunity cost) · what does one more unit give me (marginal) · is this claim positive (testable) or normative (a value judgment)?*

Key concepts / "modes" to use

- **Scarcity** — limited resources vs unlimited wants; the reason economics exists.

- **The three principles: Optimization · Equilibrium · Empiricism.**
- **Opportunity cost** — the value of the *best alternative* you give up.
- **Marginal thinking** — decide by the value of *one more* unit; **law of diminishing marginal returns**.
- **Model** — a simplified representation of relationships between variables ("all models are wrong, but useful").
- **Positive vs normative** — a *testable* cause-effect claim vs a *value judgment*.

Section-by-section main points

What is economics?

Core: economics = **the study of how people allocate *scarce* resources over competing needs** — "people (try to) do the best they can for themselves with what they have."

- **Scarcity of two things:** the **things we want** (goods, services, jobs, education, relationships, recognition…) *and* the **means to get them** (money, **time, attention**, information, patience, willpower). Air to breathe isn't scarce → not an economic question.
- Every decision is shaped by **scarcity** (can't have it all), **other people** (you're not an island), and **the rules of the game** (you can't do whatever you want).
- **How societies allocate: central planners** (someone at the top decides who gets what — parents, government) vs **markets** (value set by supply & demand — from job markets to dating markets). Economics gives tools to predict how we decide and when markets do (or don't) allocate *efficiently*.

The three principles of economics

Core: the whole course rests on three ideas.

Principle	Meaning
① **Optimization**	We choose the **best feasible option** given our limited resources (money, time, attention, willpower are all scarce).
② **Equilibrium**	A situation in which **no one would benefit** from changing their *own* behaviour, given the choices of everyone else.
③ **Empiricism**	We build **models** → models generate **predictions** → we use **data/experiments** to test (falsify) them.

Optimization — opportunity cost + marginal thinking

Core: to optimise, use two tools — **opportunity cost** (weigh each use against its best alternative — "apples to apples") and **marginal thinking** (what does *one more* unit give me?).

Opportunity cost = **the value of the best alternative you give up** — nothing "free" is free if it uses a scarce resource.

OPPORTUNITY COST

$$\text{OC(action)} = \text{benefit(action)} - \text{value of the best alternative given up}$$

- *Social media:* 1 hr/day on TikTok isn't free — you give up sport, study, work. Monetise the time with the hourly wage (avg **118 DKK** in Denmark):

YEARLY TIME-COST OF 1 HR/DAY ON TIKTOK

$$118\ \text{DKK/hr} \times 1\ \text{hr/day} \times 365\ \text{days} = \mathbf{43{,}070\ DKK\ /\ year}$$

- *Trip to Berlin (worked example):* drive = 1,280 DKK & 14 h; fly = 1,930 DKK & 3 h. Flying frees **11 h** you could work:

OPPORTUNITY COST OF DRIVING INSTEAD OF FLYING

$$\text{OC(drive)} = (1{,}930 - 1{,}280) - (118 \times 11) = 650 - 1{,}298 = \mathbf{-648\ DKK} \implies \mathbf{fly}$$

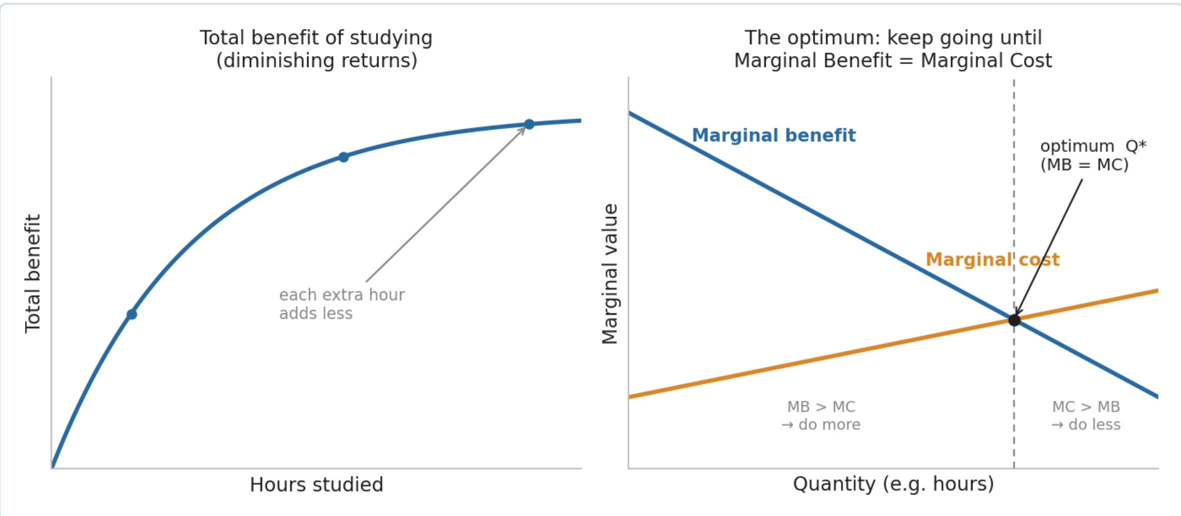
(Still a model — what if you enjoy the drive?)

Marginal thinking: decide by the value of the **next** unit — keep going while marginal benefit exceeds marginal cost; stop where they meet. That point is the **optimum**:

OPTIMUM CONDITION

$$\text{choose the quantity where}\ \ \mathbf{Marginal\ Benefit = Marginal\ Cost}\ \ (\text{MB} = \text{MC})$$

- *Night before the exam:* total benefit of studying rises then flattens; the **marginal benefit of each extra hour falls** (**law of diminishing marginal returns**). Optimal hours = where MB of one more hour just equals its MC.

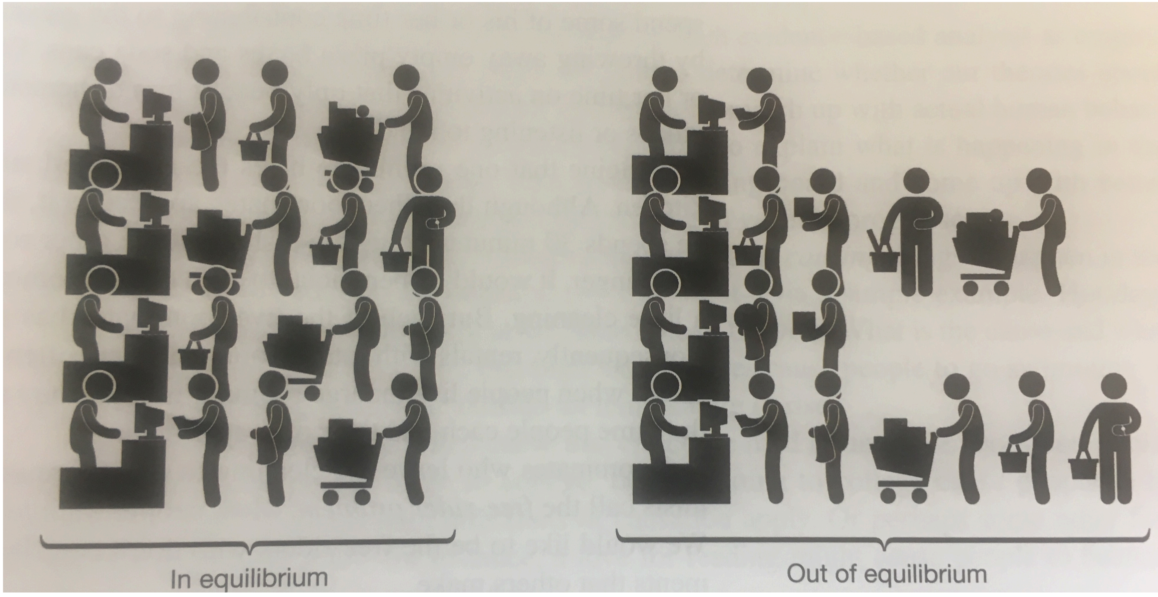


How to read it (illustrative diagram — the graph the book draws in a later chapter). *Left:* **total benefit** rises but each extra hour adds less — the curve **flattens** (diminishing returns). *Right:* that flattening means **marginal benefit slopes down**; **marginal cost** slopes up. Left of the crossing, **MB >**

MC so one more unit is worth it (*do more*); right of it, **MC > MB** (*do less*). The best you can do is the quantity where they meet — **the optimum Q*, where MB = MC.**

Equilibrium

Core: an **equilibrium** is a resting point — given what everyone else is doing, **no individual can make themselves better off by changing their own behaviour.** Market prices are the classic example (later lectures): they adjust until quantity supplied = quantity demanded and no one wants to move.

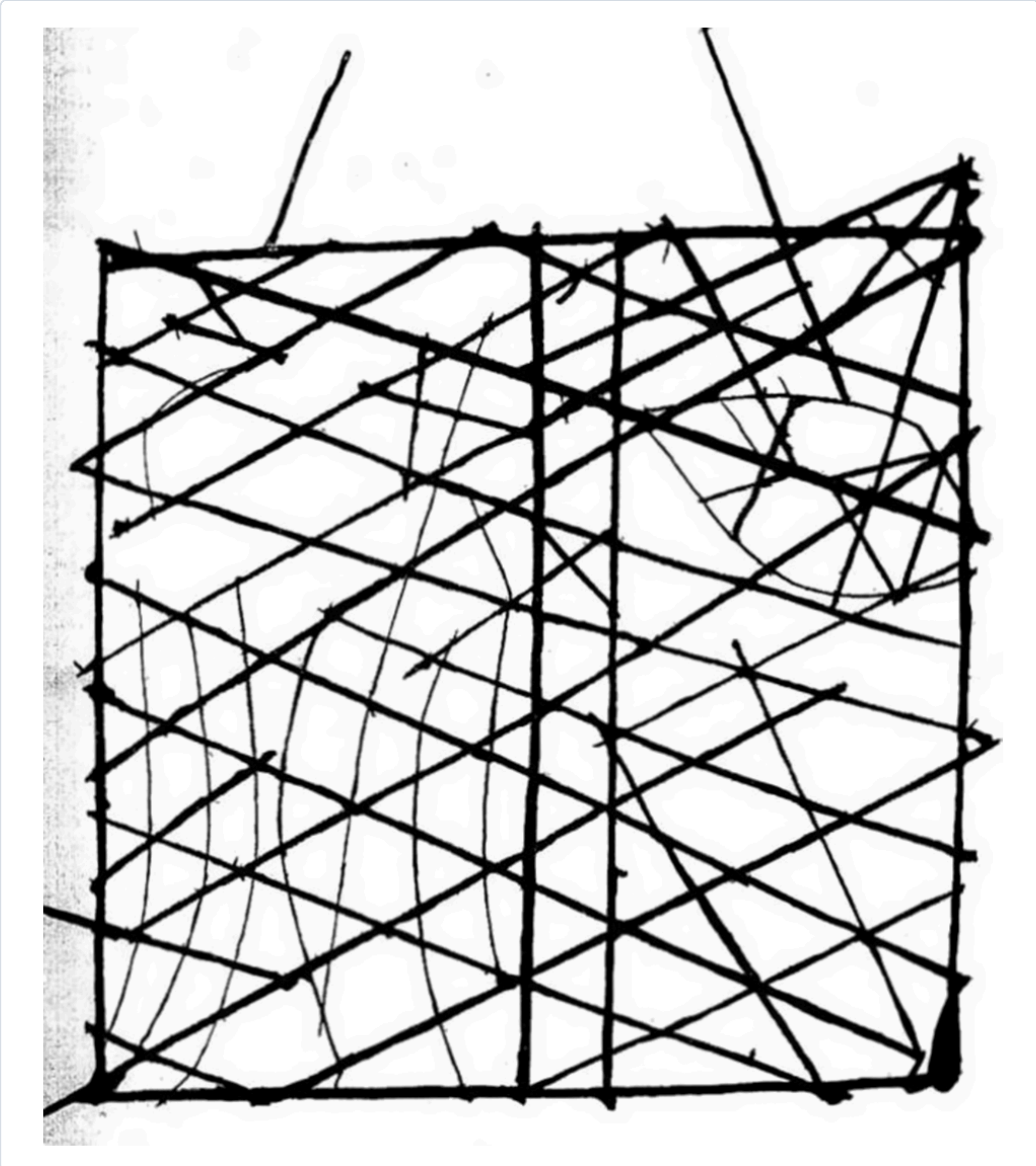


How to read it (equilibrium = supermarket queues, from the lecture). Each lane is a checkout; the queue length is how many people are waiting. **Out of equilibrium** (right) the lanes are uneven, so anyone in a long line can still gain by switching to a shorter one — people keep moving. **In equilibrium** (left) all lanes are equal, so **no one can do better by changing lane on their own** — that "no incentive to move" resting point is exactly what equilibrium means (and why market prices settle where supply meets demand).

Empiricism — models, and positive vs normative

Core: economics reasons with **models** and tests them against **data** — and it deals in **positive** (testable) statements, not **normative** (value) ones.

- **A model** expresses relationships between variables and makes **simplifying assumptions** (abstracts from reality). "*All models are useful, but all models are wrong*" — the art is choosing the right model (Keynes).



How to read it (Friberg Fig. 1.1 — what a "model" is). This is a Polynesian **stick chart**: sticks map the ocean's swell patterns, shells mark islands. It's not a literal picture of the Pacific — it **strips reality down to just the features you need to navigate.** That's the book's metaphor for an economic model: deliberately simplified, "wrong" as a full description, yet **good enough to steer decisions by.**

- **Positive vs normative:**
  - **Positive** = a **testable/falsifiable** cause-effect claim — "*each additional Starbucks in a ZIP code is associated with a 0.5% rise in housing prices*" (Glaeser et al. 2018).
  - **Normative** = a **value judgment** — "*gentrification is bad because it forces the poor out.*"
  - **Economics is about positive statements**, not normative ones.
- **Correlation ≠ causation:** in summer people eat more ice-cream *and* drown more — ice-cream doesn't cause drowning (a hidden cause: heat). A theory is only useful if it's **empirically falsifiable**.

- **Expanding models to fit data:** "people only give to charity because they care about others' well-being" fails the data (public spending doesn't crowd out giving 1-for-1 — Andreoni 1990; people give to *inefficient* charities). So we add "**joy of giving**" (utility from the act of giving itself).

Friberg's framing — allocation mechanisms (Ch. 1's example)

Core: a scarce good (e.g. **university places**) can be allocated many ways — and each has different effects on **efficiency** and **equity**. This is the book's opening illustration of "the role of economics."

- **Non-price mechanisms:** aptitude tests/interviews · secondary-school grades · open entry + hard exams · **waiting lists** · **queues** ("camp on the sidewalk") · **lotteries** · social class/legacy.
- **Price mechanisms:** everyone pays the **same fee** (set high enough to clear demand) · **auction** places to the highest bidders.
- Real systems **combine** several (fees + tests + interviews). Micro gives the tools to analyse the **trade-offs** (who gets in, how tied to parents' income, effect on effort).

Cases & examples

Each: *the example → the concept it teaches.*

- **1 hr/day of TikTok** → **opportunity cost** (≈43,070 DKK/yr at the average wage) — nothing using scarce time is free.
- **Drive vs fly to Berlin** → opportunity cost done properly: include the *value of time*, and driving turns out to *cost* 648 DKK more.
- **Studying the night before an exam** → **marginal thinking** + **diminishing returns**; optimise where MB = MC.
- **Ice-cream & drowning** → **correlation ≠ causation.**
- **Starbucks & house prices** (Glaeser 2018) → a **positive** statement; **gentrification** "**is bad**" → a **normative** one.
- **Charitable giving** (Andreoni 1990) → models must be **expanded to fit data** ("joy of giving").
- **University admissions** (Friberg) → **allocation mechanisms** (price vs non-price) and their equity/efficiency trade-offs.

Key terms

Term	Meaning
Scarcity	Limited resources vs unlimited wants — the basis of economics
Optimization	Choosing the best feasible option given constraints
Equilibrium	No one gains from changing their own behaviour, given others'
Empiricism	Build models → predict → test with data
Opportunity cost	Value of the best alternative given up
Marginal thinking	Deciding by the value of one more unit
Law of diminishing marginal returns	Each extra unit adds less than the last
Model	Simplified representation of relationships between variables
Positive statement	A testable/falsifiable cause-effect claim
Normative statement	A value judgment (good/bad, should/ought)
Allocation mechanism	Rule that decides who gets a scarce good (price/lottery/queue…)

Exam pointers

- State the **three principles** (Optimization · Equilibrium · Empiricism) and define each.
- **Compute an opportunity cost** properly (include the value of time) — the Berlin example is the model answer.
- Use **marginal thinking** + **MB = MC** to find an optimum, and name the **law of diminishing marginal returns**.
- Classify a statement as **positive vs normative**, and remember **economics = positive**; watch for **correlation ≠ causation**.
- Explain why economists use **models** ("all models are wrong, but useful") and give an **allocation-mechanism** trade-off (Friberg's university example).

★ Fun facts & memorable details

> Sticky lines from Lecture 1.

- Keynes: "*Economics is a science of thinking in terms of models joined to the art of choosing models which are relevant to the contemporary world.*"
- The course mantra: "***All models are useful, but all models are wrong.***"
- **1 hour of TikTok a day ≈ 43,070 DKK a year** in forgone wages (at Denmark's ~118 DKK average hourly wage).

- Driving to Berlin *looks* cheaper (1,280 vs 1,930 DKK) but, once you price the 11 lost hours, it actually **costs 648 DKK more** than flying.
- "*During the summer people eat more ice-cream and also drown more*" — the classic **correlation-≠-causation** trap (the hidden cause is heat).
- "**Joy of giving**" (Andreoni 1990): we don't donate only for others' welfare — we get utility from the *act* of giving, which is why public spending doesn't crowd out private giving 1-for-1.
- Learning micro is "*a first course in a new language*" — a bit hard, not immediately intuitive, needs practice.

Lecture 2: Supply and Demand (Chapter 2)

Required reading: Friberg, ch. 2 (*Supply and Demand*). · Lecturer: **Luigi Butera.**

Theme of the lecture (Butera): build **tractable models of demand and supply**, then use them to find the **market equilibrium** — the price at which "*no one would benefit from changing their own behaviour given the choices of others.*" The whole of micro is understanding that equilibrium: what sets prices and quantities, and how they move.

> ★ **The heart of the chapter:** a market is one picture — a **downward-sloping demand curve** and an **upward-sloping supply curve** that cross at **exactly one point, the equilibrium (p*, q*)**, where **quantity demanded = quantity supplied**. Three skills: ① write demand & supply as **functions**; ② set them equal to **solve for p* and q***; ③ **shift** a curve and read off the new equilibrium. *Price is set by BOTH blades of the scissors (Marshall) — never demand or supply alone.*

The red thread. For *every* question ask the one diagnostic that trips people up: **is this a *movement ALONG* a curve (only the good's *own price* changed) or a *SHIFT of* the whole curve (something *else* changed — income, other prices, costs, tastes, tech)?** Own price → slide along. Anything else → the curve moves. Get that right and the rest is algebra.

Key concepts / "modes" to use

- **Demand function** `Q_D = D(p, pₛ, p_c, Y, τ)` → its 2-D **demand curve**; **inverse demand** (p as a function of q).
- **Supply function** `Q_S = S(p, p_o)` → **supply curve**; **inverse supply**.
- **Movement along vs shift** of a curve (the core distinction); **shifters** of each curve.
- **Substitutes / complements**; **normal / inferior** goods; **Giffen** good (the rare exception).
- **Market equilibrium** `Q_D = Q_S` → solve for **p*, q***; **excess demand (shortage) / excess supply (surplus)**.
- **Comparative statics** (shifts) & how the **slope** of the other curve sizes the effect.
- **Price ceiling / floor**; **perfect competition** assumptions; **Marshall's scissors**.

Section-by-section main points

2.1 Demand — the demand function

Core: demand answers "*how much do buyers want at each price?*" The **law of demand:** all else equal, **lower price → higher quantity demanded** (the curve slopes **down**).

- *In plain terms — why it slopes down:* two reasons. ① **Scarcity** — you don't have infinite money, so a lower price lets you afford more. ② **Diminishing marginal utility** — the *first* slice of pizza is worth a lot to you; the 30th almost nothing, so you'll only buy more if it's cheaper.
- **The demand function** describes quantity demanded `Q_D` as a function of price **and everything else**:

DEMAND FUNCTION

$$Q\_D = D(\ p\ ,\ p_s\ ,\ p\_c\ ,\ Y\ ,\ \tau\ ) \text{ — } p = \text{own price} \cdot p_s = \text{price of a } \textbf{substitute} \cdot p\_c = \text{price of a } \textbf{complement} \cdot Y = \text{income} \cdot \tau = \text{taxes}$$

- **How the 2-D curve appears — "flush" the other variables into a constant.** A demand *curve* plots `Q_D` against **own price only**, so you **fix** everything else at set numbers; they collapse into the intercept. The book's worked reduction — start from a specific form and plug in temperature T=20°, substitute price p_r=2, income I=5:

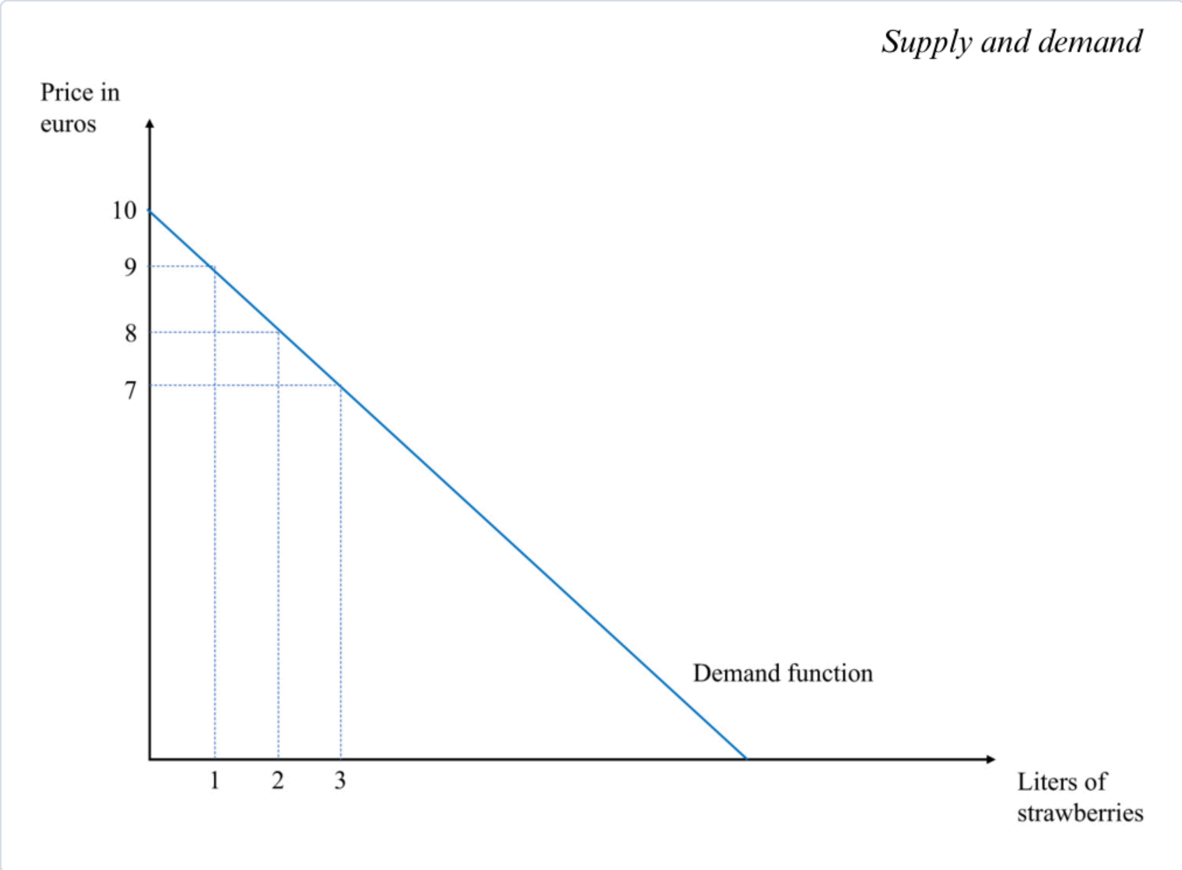
COLLAPSING TO A LINEAR DEMAND CURVE

$$Q\_D = 1 - p + 0.25{\cdot}T + 0.75{\cdot}p_r + 0.5{\cdot}I = 1 - p + \mathbf{5 + 1.5 + 2.5 = 10 - p}$$
$$\text{(intercept a = 10 bundles all the non-price factors)}$$

LINEAR DEMAND & ITS INVERSE (FOR GRAPHING)

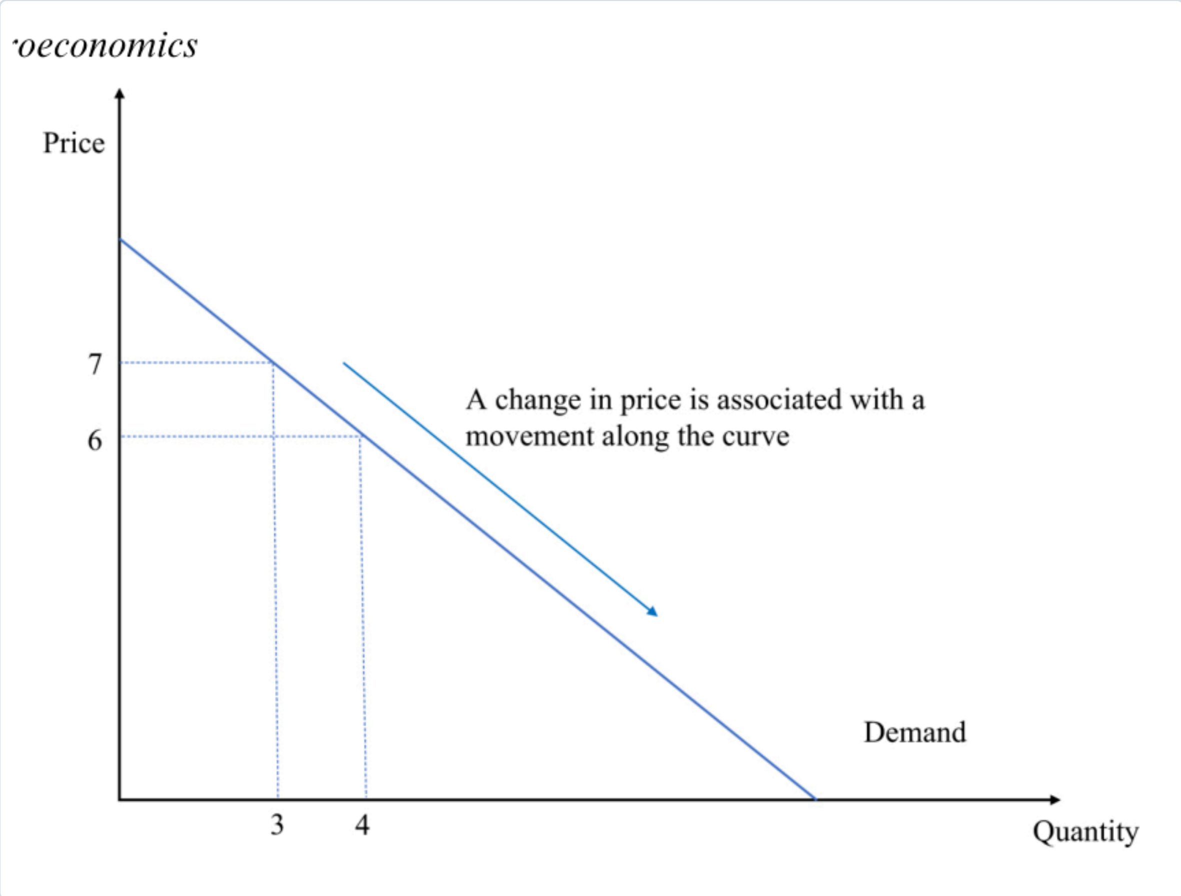
$$Q\_D = a - b{\cdot}p \rightarrow \text{e.g. } \mathbf{Q\_D = 10 - p}\ (a = 10,\ b = 1). \text{ Solve for p} \Longrightarrow \textbf{inverse demand: } \mathbf{p = 10 - Q\_D} \cdot \textit{(deck's version: } Q\_D = 20 - 2p \Longrightarrow p = 10 - \tfrac{1}{2}{\cdot}Q\_D)$$

Check: p = 7 → Q_D = 3; p = 6 → Q_D = 4. **a** = the intercept (shift it and the whole curve moves); **b** = how many units demand falls per +1 in price.



How to read it (the demand curve). By convention **price is on the vertical axis, quantity on the horizontal** (blame Alfred Marshall). Each point answers *both* questions: "at price p, how much is bought?" *and* "for quantity q, what's the most a buyer will pay?" Here the line hits **p = 10 when q = 0** (nobody buys above €10) and slides down to more litres as price falls.

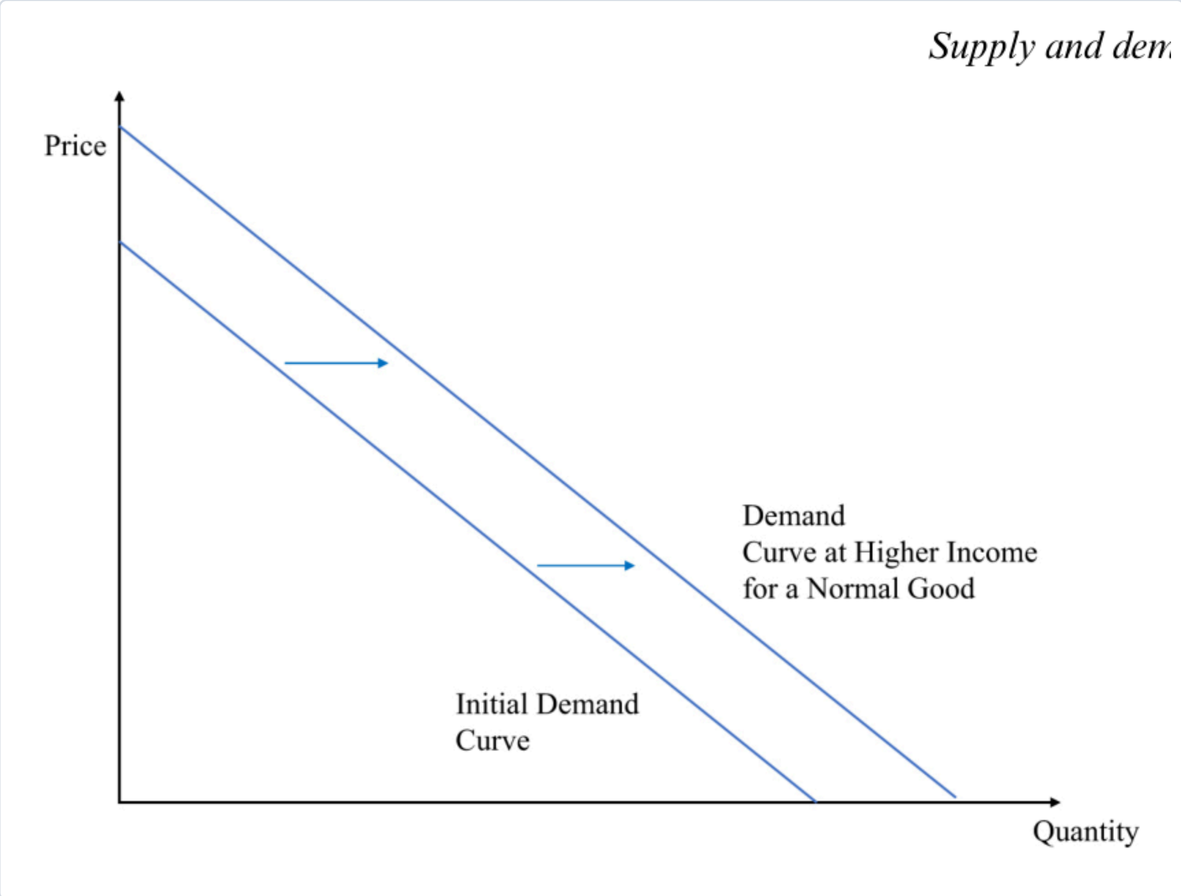
- **★ Movement ALONG vs SHIFT — the make-or-break distinction:**
  - **Own price changes → MOVE ALONG** the existing curve (you slide to a new point).
  - **Anything else changes → the whole curve SHIFTS** (a *new* curve).

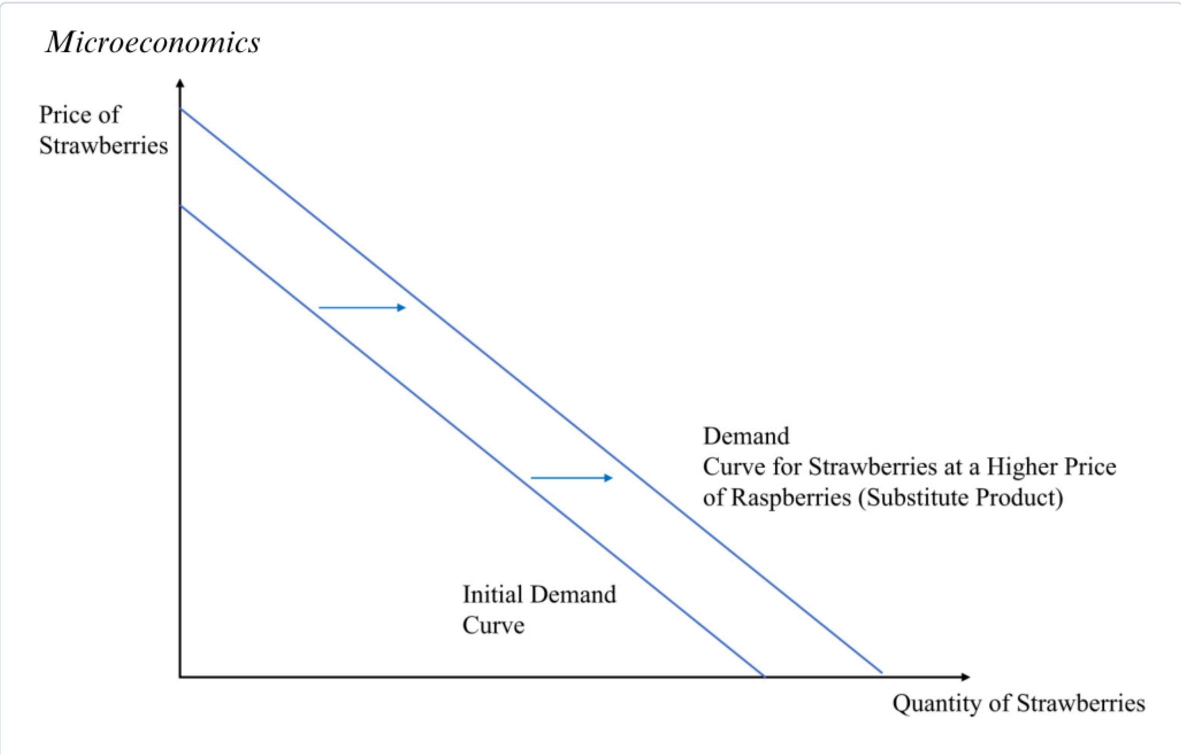


How to read it (movement along). Only the **own price** moved (7→6), so you **stay on the same curve** and just slide to a new point (q 3→4). No new curve is drawn. Contrast the shifts below, where the curve itself jumps.

- **Demand shifters (each moves the *whole* curve = changes the intercept a):**

Shifter	Direction	Why
Income ↑ (normal good)	shift **out** (right)	richer → buy more at every price
Income ↑ (inferior good, e.g. instant noodles)	shift **in** (left)	richer → buy *less* of it
Price of a substitute ↑ (raspberries ↑ → strawberry demand)	shift **out**	switch toward the now-relatively-cheaper good
Price of a complement ↑ (cereal ↑ → milk demand)	shift **in**	the pair is consumed together
Tastes / expectations (heatwave → ice-cream; COVID → toilet paper)	out or in	common-sense direction





How to read them (shifts). The whole line moves **right (out)** = more demanded at *every* price. Fig 2.4: higher **income** (normal good). Fig 2.6: a **substitute** (raspberries) got dearer, so buyers swing to strawberries. Note you are *not* sliding along the old curve — a brand-new curve sits to the right.

- **Substitutes vs complements (definitions): substitutes** — demand for good 1 **rises** when good 2's price rises (raspberries ↔ strawberries). **Complements** — demand for good 1 **falls** when good 2's price rises (milk ↔ cereal).

- **From individual to aggregate demand (deck).** Market demand = **horizontal sum** of every buyer's demand — *add the quantities at each price*:

> AGGREGATE (MARKET) DEMAND
>
> $$Q(p) = q_1 + q_2 + \ldots = D_1(p) + D_2(p) + \ldots \textit{ (add quantities across buyers at the same price)}$$

Deck example: Mike's inverse demand $p = 100 - 2q_M$, Linda's $p = 100 - \frac{1}{2}q_L$ → invert each to $q(p)$, then add: at each price, total $q = q_M + q_L$. *(Watch for a **kink** where one buyer drops out at high prices.)*

- **The T-shirt trap (don't confuse products):** seeing pricey Gucci shirts outsell cheap generics is **not** an upward-sloping demand curve — those are **different goods**. A demand curve compares a good **to itself** (Gucci cutting *its own* price sells *more*). Real upward-sloping demand = the very rare **Giffen good** (Irish-famine potatoes; poor Chinese rice households — Jensen & Miller 2008).

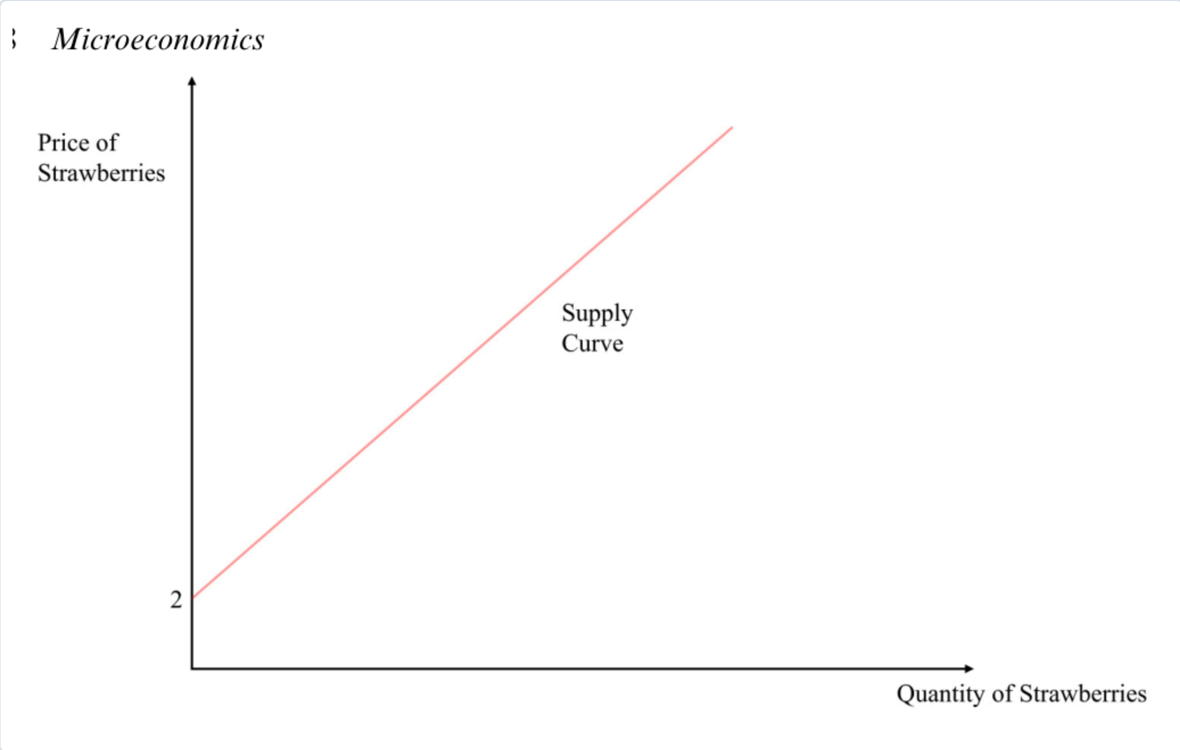
2.2 Supply — the supply function

Core: supply answers "*how much do firms bring to market at each price?*" Firms are **price-takers** (they choose *quantity*, not the price). The curve usually slopes **up** — higher price → more supplied.

- *In plain terms — why upward:* a higher price (i) makes it worth producing more even as costs rise, and (ii) draws in higher-cost producers who couldn't profit at the low price.
- **The supply function** (quantity supplied depends on price and input/other factors):

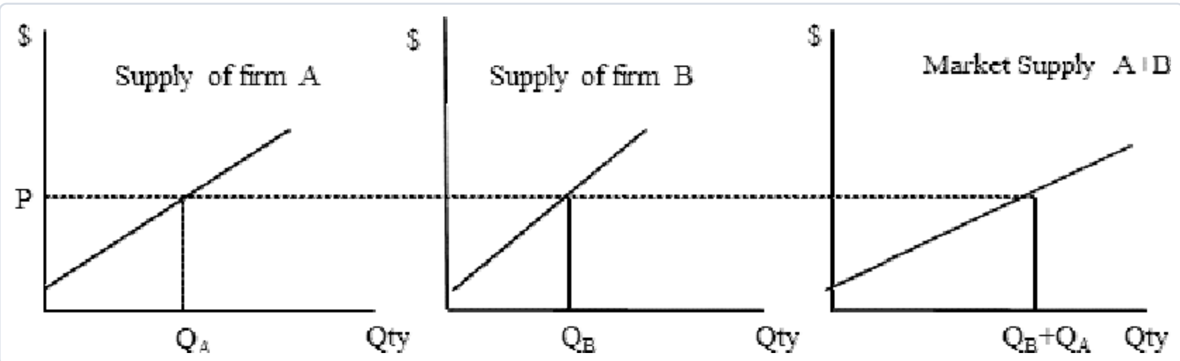
> SUPPLY FUNCTION & ITS INVERSE
>
> $$Q_S = S(\ p\ ,\ p_o\ ) \text{ — } p_o = \text{price of inputs. Book's linear example: } \mathbf{Q_S = -2 + p} \implies \text{inverse supply } \mathbf{p = 2 + Q_S}$$

- **Why the "−2" (a negative intercept that isn't weird):** read it in **inverse** form — **price must clear €2 before *any* is supplied** (at $p = 2$, $Q_S = 0$). *Check:* $p = 3 \to Q_S = 1$; $p = 4 \to Q_S = 2$.



How to read it (the supply curve). Start at the **€2 price-intercept** (below it, zero supply) and climb: each extra euro of price brings one more litre to market. Like demand, an **own-price** change = **move along**; anything else = **shift**.

- **Supply shifters (move the whole curve): input/cost ↑** (wages, transport) → shift **in** (need a higher price for the same quantity); **cheaper inputs / new tech / more sellers / market opens to foreign suppliers** → shift **out**; **disruptions** (strikes, extreme weather, mine closures) → shift **in**.
- **From individual to aggregate supply (deck):** market supply = **horizontal sum** of each firm's supply — add quantities across firms at each price.



How to read it (aggregate supply). At any price **P**, read each firm's quantity off its own curve (Q_A, Q_B) and **add them across** — the market curve is the sum, so it's **flatter** (more responsive) than any single firm's.

★ 2.3 Market equilibrium — solving for p* and q*

Core: equilibrium = the **one price where quantity demanded = quantity supplied** (`Q_D = Q_S`), the **market-clearing** price — no shortage, no glut, nobody wants to move. **This is the calculation to master.**

- **The recipe (memorise these three steps):**

STEP 1 — SET DEMAND = SUPPLY

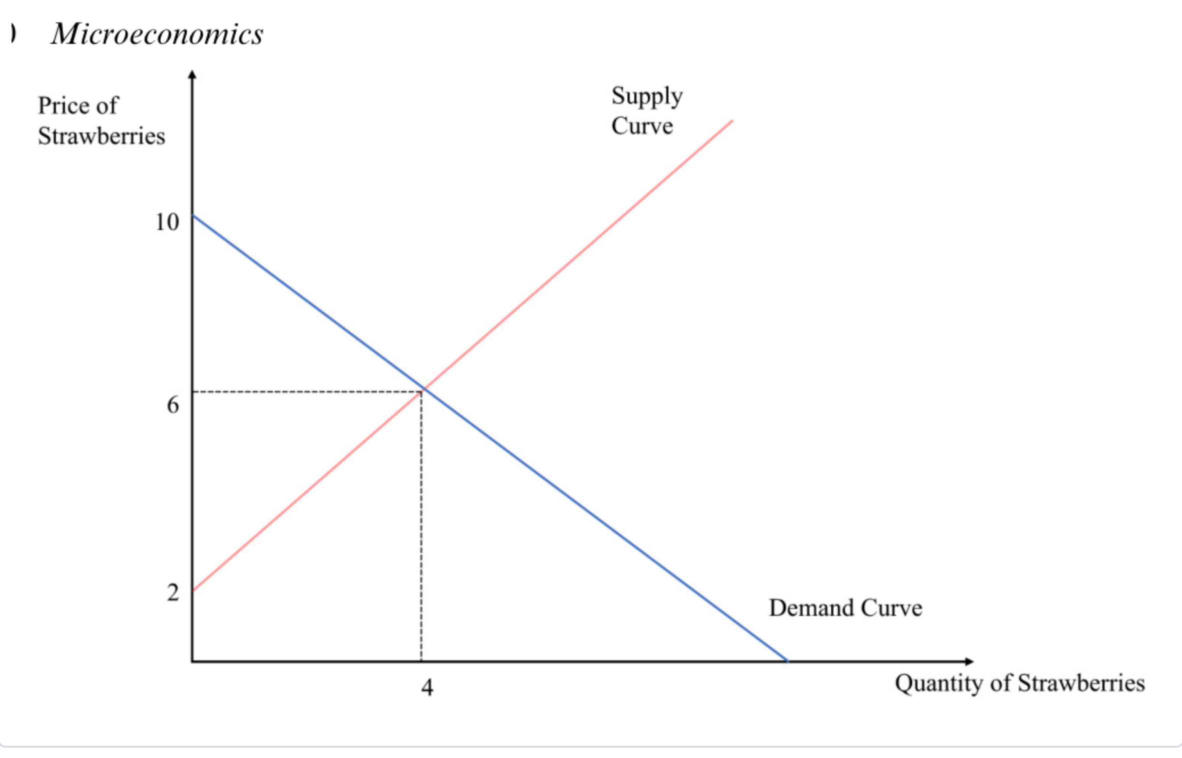
$$Q\_D = Q\_S \implies 10 - p = -2 + p$$

STEP 2 — SOLVE FOR THE EQUILIBRIUM PRICE P*

$$10 + 2 = p + p \implies 12 = 2p \implies \mathbf{p^* = 6}$$

STEP 3 — PUT P* BACK INTO EITHER CURVE FOR Q*

$$Q^* = 10 - 6 = \mathbf{4} \text{ (check: } Q\_S = -2 + 6 = 4 \checkmark) \implies \textbf{equilibrium } \mathbf{(p^*, q^*) = (6, 4)}$$



How to read it (equilibrium). The **crossing point** is the only place the two plans agree — buyers want exactly what sellers offer (4 litres at €6). Everywhere else, one side is frustrated and price gets pushed toward the cross.

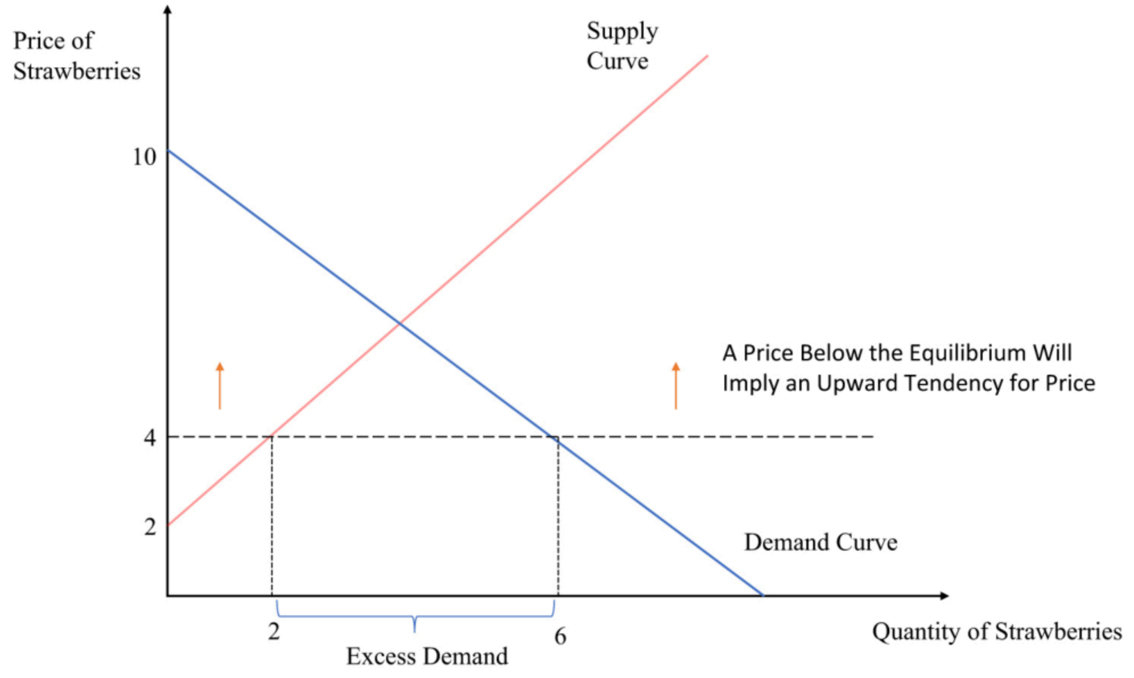
- **Off-equilibrium: excess demand & excess supply** (this is also *exactly* how you handle price controls below):

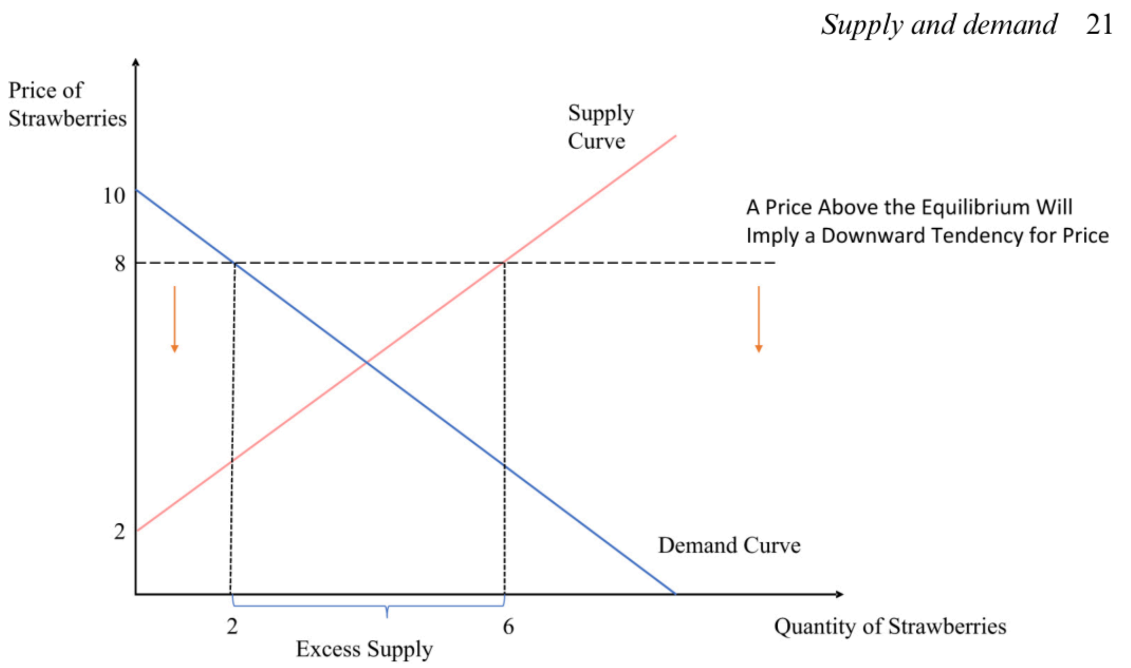
EXCESS DEMAND (SHORTAGE) — PRICE BELOW P*

$$\text{at } p = 4\text{: } Q\_D = 10–4 = \mathbf{6},\ Q\_S = -2+4 = \mathbf{2} \implies \text{shortage} = Q\_D - Q\_S = \mathbf{4} \to \text{upward pressure on price}$$

EXCESS SUPPLY (SURPLUS) — PRICE ABOVE P*

at $p = 8$: $Q\_D = 10-8 = \mathbf{2}$, $Q\_S = -2+8 = \mathbf{6} \implies$ surplus $= Q\_S - Q\_D = \mathbf{4} \rightarrow$ downward pressure on price





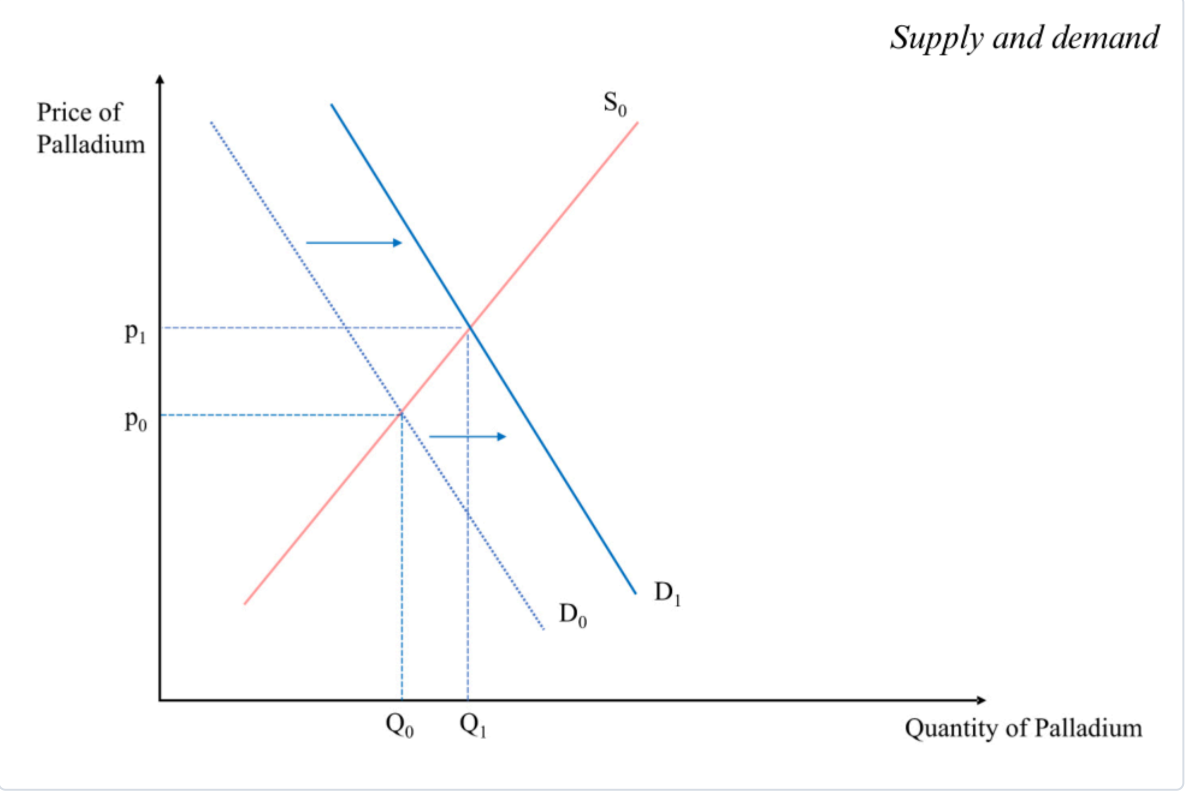
How to read them. Pick the off-equilibrium price, drop a vertical line, and read **both** curves: the **horizontal gap** between them is the shortage (demand side wider) or surplus (supply side wider). The gap is what pushes price back toward p*.

- **Marshall's scissors:** asking whether *demand or supply* sets the price is like asking *which blade of a scissors cuts* — **both do.** This dissolves the **diamond–water paradox**: water is cheap not because it's little wanted but because it's **abundant in supply**; diamonds are dear because they're **scarce**.
- *(Who actually "sets" the price if everyone's a price-taker? The theoretical* ***Walrasian auctioneer*** *— and* ***Vernon Smith's 1962 experiments*** *showed real people reach the equilibrium price fast.)*

2.4 Shifts in the equilibrium — comparative statics

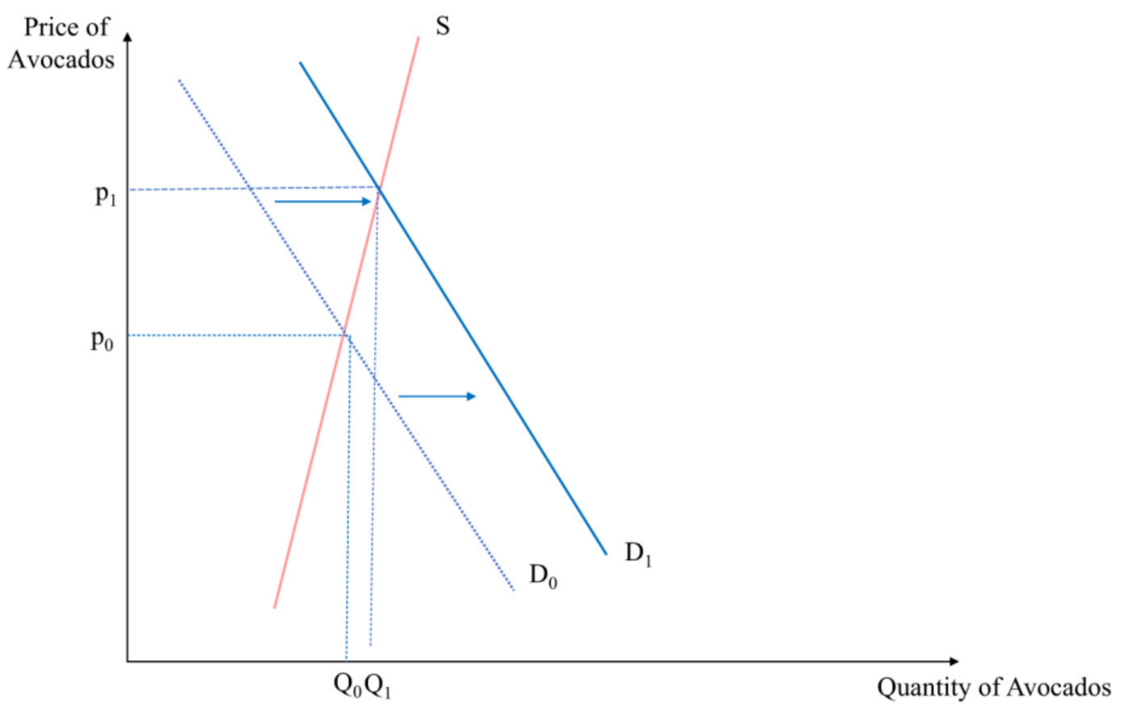
Core: to analyse any shock, **don't jump to price/quantity.** First figure out **which curve moves and which way**; the effect on p* and q* follows.

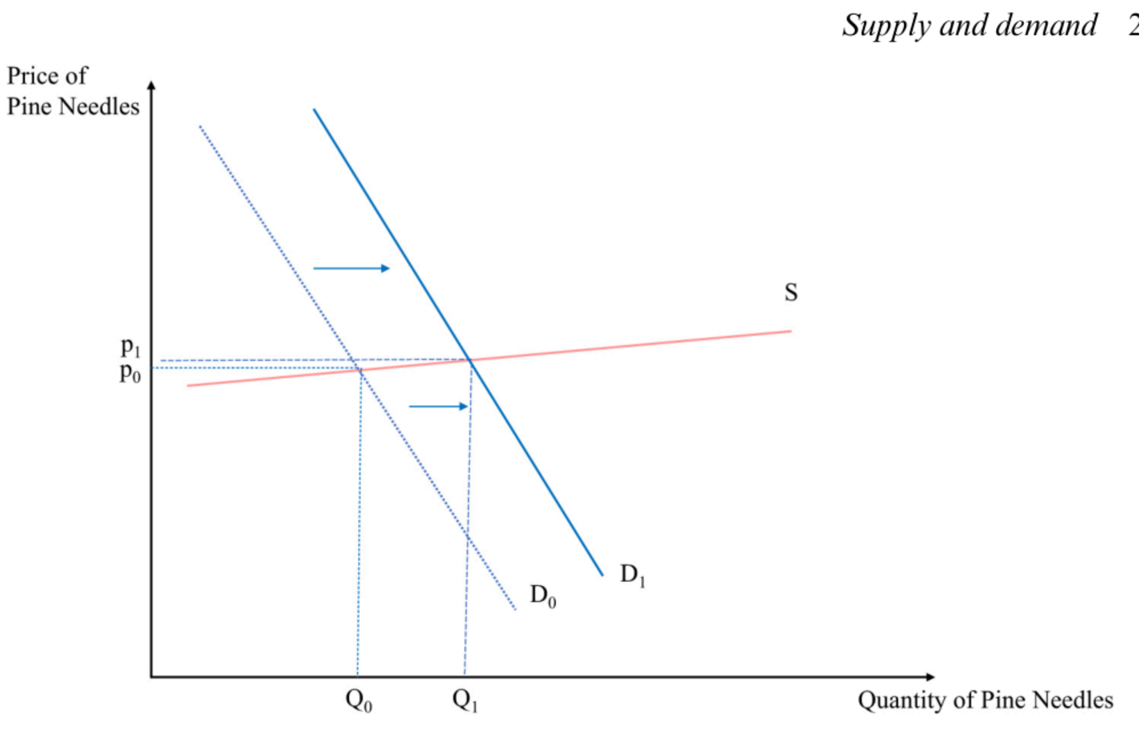
- **The method (Butera/Marshall — one change at a time, *ceteris paribus*): ① Demand or supply?** (falling incomes/other-good prices → demand; input costs/disruptions → supply). ② **Which direction** does it move *at a given price*? ③ *Then* read the new crossing. ④ **Reminder:** a *shift* is not a movement along — so a demand shift giving **both** higher price **and** higher quantity does **not** break the law of demand.



How to read it (a demand shock). Demand shifts **out** ($D_0 \to D_1$); supply is unchanged, so producers **move along** supply to the new cross — **both p and q rise**. (Add a simultaneous supply **fall** and price rises *unambiguously*, but the **quantity effect becomes ambiguous** — it depends which shift is bigger.)

- ★ **Slope decides the *size* of the effect.** A **demand** shock's impact depends on the **slope of the *supply* curve**:





How to read them. Same outward demand shift, opposite results. **Steep supply** (avocados — trees take years, stock is fixed short-run): price jumps, quantity barely moves. **Flat supply** (pine needles — anyone can gather more): quantity jumps, price barely moves. *The flatter the other curve, the more a shock shows up as **quantity**; the steeper, the more it shows up as **price**.*

Price ceilings & floors — when the government fixes the price (deck)

Core: a **binding price control** stops the market clearing, and you find the resulting gap with the **exact excess-demand/supply calculation above** — just set p to the controlled price.

- **Price ceiling** = a legal **maximum** (e.g. **rent control**). To bind it must be **below p***, so **Q_D > Q_S → a shortage** (queues, waiting lists, black-market resale). *Compute:* plug the capped p into both curves; shortage = Q_D − Q_S.
- **Price floor** = a legal **minimum** (e.g. the **minimum wage** in the labour market). To bind it must be **above p***, so **Q_S > Q_D → a surplus** (unsold goods; in the labour market, **unemployment** — more people want to work than firms will hire). *Compute:* surplus = Q_S − Q_D.
- **The trade-offs (positive, not normative):** *ceilings* — **pro:** keep essentials affordable for the less-well-off; **con:** shortages, quality decline, and markets "find a way around" (rent control can even be **regressive**; Venezuela's caps produced empty shelves). *floors* — **pro:** a living wage; **con:** firms substitute **capital for labour** (self-checkout at Føtex), cut hours, or make jobs temporary. "*Good economics generates **positive** statements — evidence on costs and benefits — to inform the **normative** political choice.*"

Is the model realistic? — perfect competition (deck)

Core: the supply-demand model assumes **perfect competition** — "*every model is wrong, but useful.*"

- **Assumptions:** ① every agent is a **price-taker**; ② goods are **homogeneous** (perfect substitutes — one farm's wheat = another's; Shell petrol = Exxon petrol); ③ **perfect information** about prices/quality; ④ **low transaction costs**; (and many buyers & sellers). **Break these** and you need refinements — **monopoly, market failures** (later chapters).

Cases & examples

Each: *the example → the concept it teaches.*

- **Strawberries (the running example)** → the whole toolkit: `Q_D = 10 − p`, `Q_S = −2 + p`, equilibrium **(6, 4)**, and the €4/€8 shortage/surplus.
- **Diamond–water paradox** → price is set by **supply *and* demand** (Marshall's scissors), not usefulness — water's cheap because it's abundant.
- **Palladium** (tighter emission rules → catalytic-converter demand ↑; S. African mine disruptions → supply ↓) → **comparative statics** with one and two curves moving.
- **2008 world rice spike & Haiti riots** (India's export ban cut supply; panic stockpiling shifted demand out) → a real **double shift**; prices later fell as stockpiles released + the financial crisis hit demand.
- **Instant noodles** → **inferior good**; **raspberries** → **substitute**; **cereal & milk** → **complements**.
- **Giffen goods** (Irish-famine potatoes; poor Chinese rice households) → the rare **upward-sloping demand** exception.
- **COVID toilet-paper stockpiling** → an **expectations** shift of demand.
- **Avocados vs Swedish pine-needle tea** → **steep vs flat supply** sizing a demand shock.
- **Rent control / Venezuela** → **price ceiling → shortage**; **minimum wage** → **price floor → surplus (unemployment)**.

Formula sheet — quick reference

Object	Formula	Note
Demand function	$Q\_D = D(p, p_s, p\_c, Y, \tau)$	many variables; curve fixes all but p
Linear demand / inverse	$Q\_D = a - b \cdot p$ / $p = (a - Q\_D)/b$	a = intercept (shifters), b = slope
Supply function / inverse	$Q\_S = -2 + p$ / $p = 2 + Q\_S$	negative intercept = min price to supply
Aggregate (market)	$Q(p) = \Sigma D_i(p)$; $\Sigma S_i(p)$	**horizontal** sum (add quantities)
Equilibrium	**set Q_D = Q_S → p*, then q***	the market-clearing point
Excess demand (shortage)	$Q\_D - Q\_S$ at $p < p^*$	→ upward price pressure; price ceiling
Excess supply (surplus)	$Q\_S - Q\_D$ at $p > p^*$	→ downward pressure; price floor

Exam pointers

- **Solve a market from scratch:** given a linear demand and supply, **set them equal**, get **p***, back-substitute for **q*** — the strawberry case (p*=6, q*=4) is the model answer.
- **Compute a shortage/surplus:** plug a below- or above-equilibrium price into *both* curves and take the gap (**Q_D − Q_S** or **Q_S − Q_D**) — and know a **ceiling→shortage**, **floor→surplus**.
- **Diagnose movement vs shift**, and for a shock say **which curve moves, which way**, and the effect on p* & q* (incl. the **ambiguous-quantity** double-shift and the **slope** point).
- Explain **why demand slopes down** (scarcity + diminishing marginal utility) and **Marshall's scissors** / the diamond–water paradox.
- Name the **perfect-competition** assumptions and why the model is still "useful though wrong."

★ Fun facts & memorable details (Lecture 2)

> Sticky bits from Supply & Demand.

- **Marshall's scissors:** asking whether demand or supply sets the price is like asking **which blade of the scissors** does the cutting — both do.
- The **diamond–water paradox** is only a paradox if you forget supply: water is life-or-death yet cheap **because it's abundant**.
- **Giffen goods** really exist: when their staple got dearer, poor Chinese rice households bought **more** rice, not less (Jensen & Miller 2008) — they could no longer afford anything else.
- A negative supply intercept (**Q_S = −2 + p**) isn't a mistake — it just means **price must top €2 before anyone supplies a drop.**
- **Vernon Smith** won a Nobel (2002) for showing that real people in a lab **converge on the equilibrium price** startlingly fast — the "Walrasian auctioneer" made flesh.

- Rent control's dirty secret: it can be **regressive** and spawn black-market "key money" — a "tale of good intentions" (cf. Venezuela's empty shelves).
- Minimum wage as a price floor nudges firms toward **capital over labour** — the **self-checkout at Føtex** is a supply-and-demand diagram in the wild.